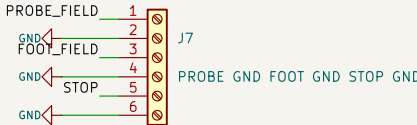
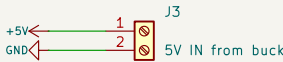
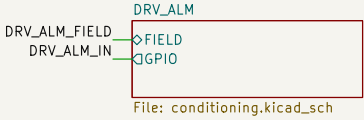
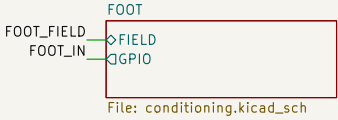
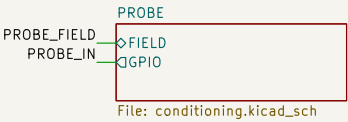
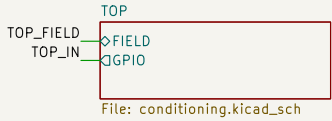
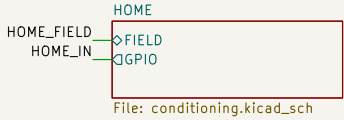
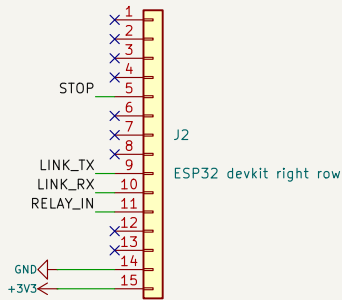
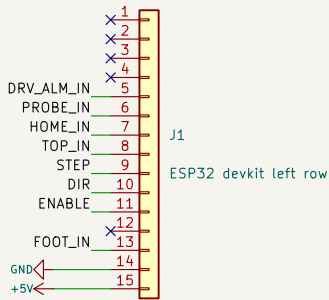
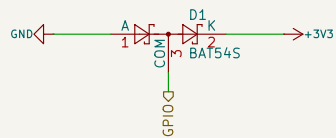
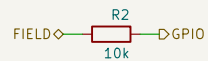


J1 pins 1–15: EN VP_36 VN_39 D34 D35 D32 D33 D25 D26 D27 D14 D12 D13 GND VIN
J2 pins 1–15: D23 D22 TX0_1 RX0_3 D21 D19 D18 D5 D17 D16 D4 D2 D15 GND 3V3
DOIT 30-pin layout – verified against the HW–394 board silkscreen, 2026–09–14.
GPIO 34 reserved (feedback), 35 = driver alarm, not configured yet.



LINK carries GND, TX, RX only. NEVER join this 3V3 to the panel's 3.3 V.
TB6600 common is +3.3 V (not 5 V). STOP goes direct to GPIO 21 (feed_hold_pin).
Sensor shields: GND terminal at this end only.





Pull-up on the WIRE side of the 10k; clamp and 100n on the GPIO side.
A pull-up on the GPIO side leaves a closed switch at ~2.2 V, which does not read LOW.
 $10k \times 100n = \sim 1 \text{ ms}$ filter (ELE-04). BAT54S makes a mis-wired PNP sensor survivable.

routerLift

Sheet: /HOME/

File: conditioning.kicad_sch

Title: Input conditioning channel (x5)

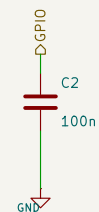
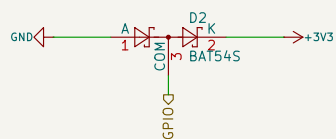
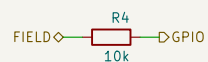
Size: A4

Date: 2026-09-13

Rev: H

KiCad E.D.A. 10.0.6

Id: 2/6



Pull-up on the WIRE side of the 10k; clamp and 100n on the GPIO side.
A pull-up on the GPIO side leaves a closed switch at ~2.2 V, which does not read LOW.
 $10k \times 100n = \sim 1 \text{ ms}$ filter (ELE-04). BAT54S makes a mis-wired PNP sensor survivable.

routerLift

Sheet: /TOP/

File: conditioning.kicad_sch

Title: Input conditioning channel (x5)

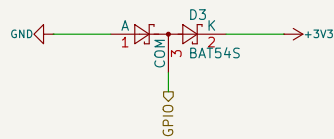
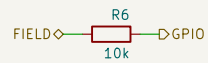
Size: A4

Date: 2026-09-13

Rev: H

KiCad E.D.A. 10.0.6

Id: 3/6



Pull-up on the WIRE side of the 10k; clamp and 100n on the GPIO side.
A pull-up on the GPIO side leaves a closed switch at ~2.2 V, which does not read LOW.
 $10k \times 100n = \sim 1 \text{ ms}$ filter (ELE-04). BAT54S makes a mis-wired PNP sensor survivable.

routerLift

Sheet: /PROBE/
File: conditioning.kicad_sch

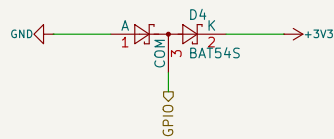
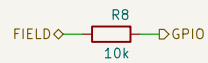
Title: Input conditioning channel (x5)

Size: A4 Date: 2026-09-13

KiCad E.D.A. 10.0.6

Rev: H

Id: 4/6



Pull-up on the WIRE side of the 10k; clamp and 100n on the GPIO side.
A pull-up on the GPIO side leaves a closed switch at ~2.2 V, which does not read LOW.
 $10k \times 100n = \sim 1 \text{ ms}$ filter (ELE-04). BAT54S makes a mis-wired PNP sensor survivable.

routerLift

Sheet: /FOOT/

File: conditioning.kicad_sch

Title: Input conditioning channel (x5)

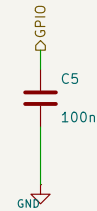
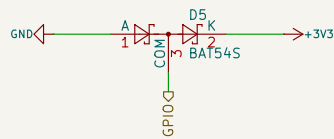
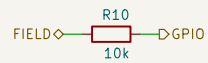
Size: A4

Date: 2026-09-13

Rev: H

KiCad E.D.A. 10.0.6

Id: 5/6



Pull-up on the WIRE side of the 10k; clamp and 100n on the GPIO side.
A pull-up on the GPIO side leaves a closed switch at ~2.2 V, which does not read LOW.
 $10k \times 100n = \sim 1 \text{ ms}$ filter (ELE-04). BAT54S makes a mis-wired PNP sensor survivable.

routerLift

Sheet: /DRV_ALM/
File: conditioning.kicad_sch

Title: Input conditioning channel (x5)

Size: A4 Date: 2026-09-13
KiCad E.D.A. 10.0.6

Rev: H
Id: 6/6